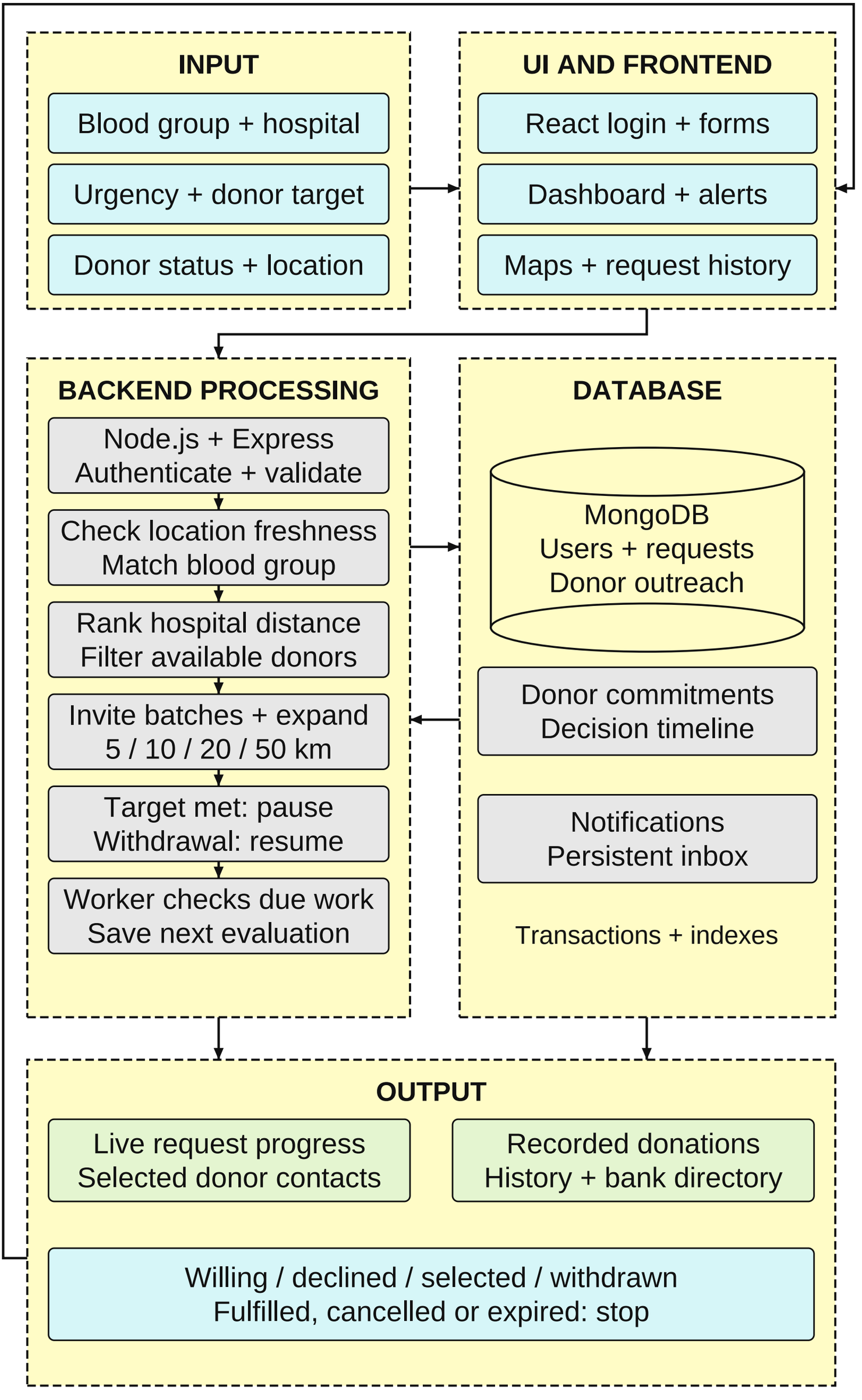
INPUT
Blood group + hospital
Urgency + donor target
Donor status + location
UI AND FRONTEND
React login + forms
Dashboard + alerts
Maps + request history
BACKEND PROCESSING
Node.js + Express
Authenticate + validate
Check location freshness
Match blood group
Rank hospital distance
Filter available donors
Invite batches + expand
5 / 10 / 20 / 50 km
Target met: pause
Withdrawal: resume
Worker checks due work
Save next evaluation
DATABASE
MongoDB
Users + requests
Donor outreach
Donor commitments
Decision timeline
Notifications
Persistent inbox
Transactions + indexes
OUTPUT
Live request progress
Selected donor contacts
Recorded donations
History + bank directory
Willing / declined / selected / withdrawn
Fulfilled, cancelled or expired: stop